AGENTIC AI CA-3

JanSetu AI

Adaptive Civic Incident Resolution

From citizen reports to verified, cleaner and safer cities.

REPORT → UNDERSTAND → ADAPT → VERIFY

Potholes
Road defects and hazards

Waterlogging
Standing water and blocked routes

Garbage
Overflow and accumulation

Academic PoC • Synthetic incidents • Simulated execution • Human-governed closure

DOMAIN-SPECIFIC PROOF OF CONCEPT

A civic incident is not resolved until the outcome is verified.

JanSetu coordinates incident understanding, risk assessment, response planning, failure recovery and independent closure checks.

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Github link:https://github.com/Harshad2321/JanSetu-AI

Civic complaints need a reliable path to resolution

Domain: civic infrastructure operations — potholes, waterlogging and garbage accumulation.

Problem statement

Unstructured reports, uncertain evidence and changing resource constraints make it difficult to prioritize incidents, plan responses and prove closure consistently.

Why this domain?

Civic incidents are location-dependent, multi-step and sometimes safety-sensitive. They require coordination, explicit decisions and evidence—not a one-turn answer.

Who benefits?

Citizens gain clearer status; officers get structured context and review gates; administrators gain traceable decisions and operational trends.

CORE PROBLEM

A complaint can be marked complete even when the plan failed, key information is missing, or evidence is insufficient.

THE DESIGN GOAL

REPORT → STRUCTURE → PLAN → ADAPT → VERIFY

Scope boundary: synthetic civic data and simulated field actions. JanSetu does not dispatch real crews or repair real infrastructure.

Agentic AI is justified because the workflow includes conditional routing, tools, persistent state, retries, human approvals and independent validation.

From fragmented handling to an auditable workflow

JanSetu is designed to improve coordination and traceability; impact must be measured in future field evaluation.

TYPICAL PROCESS RISKS	JANSETU RESPONSE
Incomplete reports Missing category, location, duration or impact.	**Quality gate** Detect missing fields or disagreement; clarify or review.
Fragmented context Nearby reports and recurring hotspots may be overlooked.	**Spatial context** Find candidate duplicates and clusters; never auto-merge.
Static planning The assigned resource may be unavailable.	**Adaptive recovery** Convert failure into constraints and create a revised plan.
Weak closure evidence A success message may not prove completion.	**Independent verification** Keep the incident open until closure criteria pass.

Expected process benefit: clearer decisions, bounded recovery, preserved history and closure that depends on evidence—not assumed success.

Six specialist agents coordinated by one orchestrator

Agents are typed logical components; not every role is an independent LLM.

CENTRAL ORCHESTRATOR • dispatches tasks • controls state transitions • enforces policy • bounds retries • records audit events

A1 Incident Intelligence

Structures the fused report; detects missing or conflicting facts; requests clarification.

A2 Geo-Spatial Intelligence

Checks distance and text similarity; flags unconfirmed duplicate candidates and hotspots.

A3 Risk & Priority

Computes an explainable rule-based score; mandatory danger rules can force escalation.

A4 Response Planning

Creates temporary-relief and permanent-resolution plans with resources and closure criteria.

A5 Adaptive Recovery

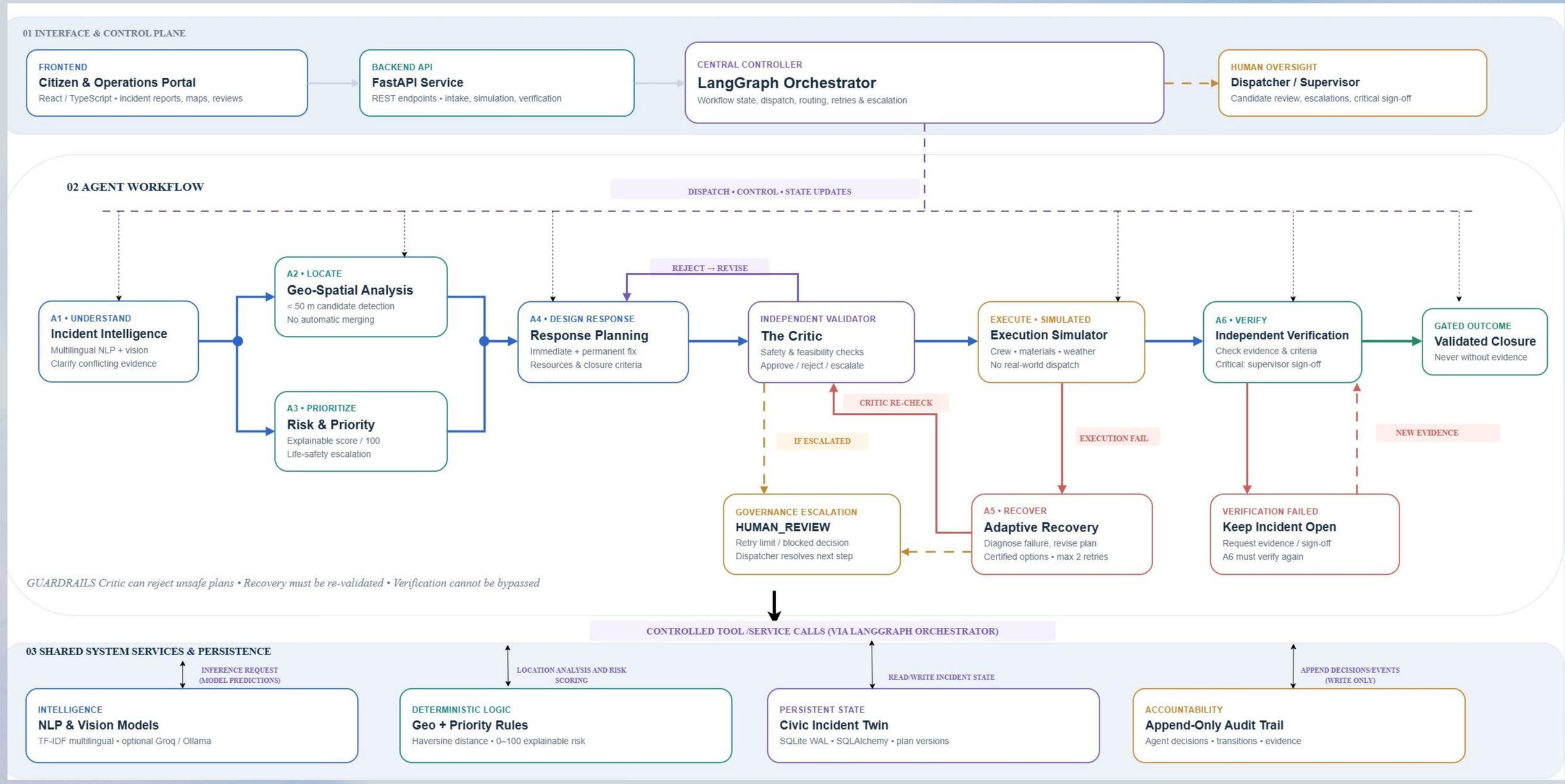
Uses failure reasons as constraints; selects a safe alternative; maximum two retries.

A6 Independent Verification

Checks completion evidence and sign-offs; approves or blocks closure independently.



Architecture Diagram



The interface should make every decision inspectable

Show the live application during the demo; this slide maps the user journey to visible outputs.

JANSETU / INCIDENT WORKSPACE

SYNTHETIC

Incident: Pothole near a busy road

CLASSIFIED | HIGH PRIORITY | SIMULATED

Incident summary

Category • location • impact
Confidence / missing fields

Plan & resources

Temporary relief + permanent plan
Assigned synthetic resources

Execution history

Attempt 1: crew unavailable
Attempt 2: alternative simulated

Verification gate

Evidence checklist
PASS → close / FAIL → keep open

Illustrative layout — replace with genuine application screenshots before submission.

LIVE DEMO SEQUENCE

1. **Submit a synthetic report**
   Text + optional image + coordinates
2. **Inspect model outputs**
   Category, confidence and fallback status
3. **Follow agent decisions**
   Priority, candidate duplicates and plan
4. **Inject a failure**
   Crew / material unavailable in simulator
5. **Verify closure**
   Missing evidence must block closure

Note: current technical overview specifies React + FastAPI.

Validation tests the workflow—not just the prediction

Results below are reported in the supplied technical overview; rerun on the current checkout before submission.

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invariant checks

Reported validation suite passed targeted workflow and safety invariants.

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backend tests

Latest supplied Git verification report; rerun to confirm current status.

Passed

Frontend build

Reported production build passed; lint had 22 warnings and zero errors.

NLP CLASSIFIER — SYNTHETIC EVALUATION

Metric	Hold-out	Challenge
Samples	60	30
Accuracy	100.0%	73.33%
Macro-F1	100.0%	73.3%

Workflow checks covered

- Closure blocked without evidence
- Simulation explicitly labelled
- Mandatory hazard escalation
- Retries bounded
- Failure history preserved
- Unsafe alternatives rejected
- Duplicates not auto-merged

Interpretation: perfect synthetic hold-out performance does not prove real-world generalization. The lower challenge score shows sensitivity to changed wording; real complaint data, multilingual robustness and confidence calibration remain unproven.

A credible PoC makes its boundaries explicit

JanSetu demonstrates a governed workflow; field deployment would require additional data, validation, integrations and approvals.

CURRENT LIMITATIONS

Synthetic inputs
Training, incidents, resources and outcomes are not representative of real city operations.

Model uncertainty
Vision may be unavailable or misclassify; reported confidence is not calibrated ground truth.

Heuristic prioritization
Risk weights are configurable rules, not an externally validated public-safety model.

No physical confirmation
Simulator outcomes and uploaded evidence cannot prove real-world safety by themselves.

Integration boundary
No live municipal dispatch, crew tracking or repair integration.

FUTURE SCOPE

Real-data evaluation
Use approved, privacy-protected incident data and independent labels.

Robust multimodal models
Evaluate image/text disagreement and multilingual performance.

Calibrated risk & confidence
Measure calibration and review safety thresholds with domain experts.

Authorized integrations
Connect municipal ticketing, maps and resource systems only with permission.

Field pilot & governance
Privacy controls, human accountability, monitoring and rollback procedures.

JanSetu AI: coordinate the response, adapt to failure, and verify before closure.